Department of Computer Science and Engineering
School of Computing, Coimbatore Campus

CAPSTONE PROJECT — FINAL DOCUMENTATION REPORT

23CSE452 · Business Analytics

AgriProcure

A Predictive and Weather-Linked Analytics Framework for Indian Mandi Commodity Prices

SARIMA forecasting, FP-Growth rule mining, price-spike classification and external validation across Indian APMC mandis

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Code and dashboards: https://github.com/aathi-11/AgriProcure

Declaration of Originality

We declare that this report, *AgriProcure: A Predictive and Weather-Linked Analytics Framework for Indian Mandi Commodity Prices*, is a genuine record of work done by Team 14 for the 23CSE452 Business Analytics capstone at Amrita Vishwa Vidyapeetham, Coimbatore. We collected the price and weather data ourselves from public sources (Agmarknet/CEDA and Open-Meteo); we did not use ready-made capstone datasets from Kaggle, UCI or similar repositories. The models (CatBoost, XGBoost, LightGBM, LSTM), the SARIMA specifications, the FP-Growth rule mining and both dashboards were written and checked by us. Generative AI tools were used only as the course policy allows (see Section 8.4), and we have checked every number and calculation reported here.

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Abbreviations

APMC	Agricultural Produce Market Committee	MAPE	Mean absolute percentage error
ARM	Association rule mining	MASE	Mean absolute scaled error
CEDA	Centre for Economic Data and Analysis	MT	Metric tonnes
CV	Coefficient of variation	PCA	Principal component analysis
DPDP	Digital Personal Data Protection (Act, 2023)	ROC-AUC	Area under the ROC curve
EDA	Exploratory data analysis	SARIMA	Seasonal ARIMA
GODL	Government Open Data License – India	SHAP	Shapley additive explanations
LSTM	Long short-term memory network	TOP	Tomato, Onion, Potato
MAE / RMSE	Mean absolute / root mean squared error	VIF	Variance inflation factor

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Abstract

Wholesale vegetable prices in India swing hard and buyers rarely get advance warning. AgriProcure asks whether supply and weather signals can supply that warning for three staple crops—tomato, onion and potato (TOP)—across 70 regulated mandis in 14 states over 52 consecutive weeks (September 2025 to September 2026). After merging Agmarknet/CEDA prices and arrivals with Open-Meteo weather, the final panel holds 11,130 weekly observations.

The work has three parts. The exploratory analysis shows that prices respond non-linearly to arrivals and follow a 13-week seasonal rhythm. In the predictive modelling stage, tuned CatBoost and XGBoost classify next-week price spikes ($>5\%$ rise) with about 80% accuracy, ROC-AUC near 0.81 and macro F1 of 0.69 on strictly chronological test folds; price momentum and arrival deficits matter most. In the advanced analytics stage, a SARIMA$(1,1,1)\times(1,0,1)_{13}$ model forecasts one to four weeks ahead with an out-of-sample MASE between 0.78 and 0.86, and FP-Growth mines 31 candidate rules, with lifts reaching 2.82. An independent replication on 1,504 mandis in 29 states reproduces the SARIMA result (MASE 0.81) and 16 of the 17 actionable rules tested. A scenario model for a 10,000 MT quarterly purchase suggests rule-triggered hedging could save about Rs. 1.85 crore in a base case.

Keywords: agricultural price forecasting, mandi prices, SARIMA, association rules, gradient boosting, weather lags, procurement.

Executive Summary

Tomato, onion and potato prices can move by more than 100% within a quarter because harvest arrivals are uneven, transport gets disrupted and monsoon shocks hit without notice. For a buyer procuring 10,000 MT a quarter, a rise of just Rs. 250 per quintal adds Rs. 2.5 crore to the bill; buying during a glut without cold storage leads to spoilage instead. AgriProcure turns market and weather data into three practical tools: a weekly spike classifier, a four-week price forecast with confidence bands, and a short rulebook of conditions that tend to come before price jumps.

- **Rain gives lead time.** Heavy rainfall (above roughly 100 mm) in growing districts tends to be followed by price rises about two to three weeks later, giving buyers a window to lock in forward contracts.
- **Forecasts beat the baseline.** SARIMA beats the seasonal-naive benchmark at every horizon tested (MASE below 1), and on the larger external dataset its error stays at 10.5–13.4% MAPE at the national level.
- **Patterns hold beyond the sample.** Expanding the study 21.5-fold to 1,504 mandis, 16 of 17 actionable rules (94.1%) still showed lift above 1.
- **The savings case is plausible.** In the base scenario, rule-triggered forward buying and cold-storage buffering saves about Rs. 1.85 crore per 10,000 MT quarter, with a benefit–cost ratio near 4 (Table 7). These figures come from a scenario model, not a live trial.

1 Introduction and Objectives

1.1 Context, scope and KPIs

Mandis are state-regulated wholesale markets where most Indian farm produce is first traded. Tomato, onion and potato—the "TOP" crops—make up over half of the vegetables households buy each day, yet their prices are among the least stable: crop cycles are short, harvests arrive in bursts, storage is patchy and weather shocks are frequent. Four groups gain from better foresight: retail and supermarket buyers, food processors who need predictable input costs, buffer-stock agencies such as NAFED, and market authorities who want transparent price discovery for farmers.

We set out to answer four questions. (1) Can weather shocks in production areas predict wholesale price spikes one to four weeks ahead? (2) Does SARIMA beat a seasonal-naive heuristic, and do its intervals help with hedging? (3) Which combinations of conditions (momentum, arrival drop, rainfall) reliably come before big price rises? (4) Do models built on a benchmark set of mandis still work on markets they have not seen? Table 1 lists the objectives and the targets we judged ourselves against.

1.2 Related work and gaps

Indian mandi markets have mostly been studied through agricultural economics. Gandhi and Namboodiri [1] examined marketing margins and price spreads for fruits and vegetables and linked volatility to spatial arbitrage frictions and intermediary margins. Sekhar [2] studied spatial market integration and found that major hubs such as Azadpur and Lasalgaon move together, while regional markets can decouple for a while after climate shocks. On the forecasting side, Jha et al. [3] applied classical ARIMA-type models to potato and onion series from a single region.

Looking across this literature, we see four gaps that this project tries to close:

1. Weather is largely ignored, even though rain and heat reach the market with a delay.
2. Forecasts are rarely backtested over several origins and horizons; single hold-out splits and in-sample fit statistics are

Table 1: Business objectives, KPIs and success targets.

Objective	Scope	KPI and baseline	Target	Section
1 Price dynamics	70 mandis, 14 states, 52 weeks	CV (volatility)	Document spatial and seasonal spread	3
2 Early warning	Spike ($>5\%$ weekly rise) at $t+1$	Macro F1, ROC-AUC; persistence F1 = 0.497	F1 ≥ 0.65, accuracy $\geq 75\%$	4
3 Forecasting	Horizons $h = 1$–4 weeks	MASE, MAPE; seasonal naive MASE = 1.00	MASE < 1, MAPE $< 15\%$	5
4 Market rules	Weather and supply triggers for price shifts	Lift (1.00 = independence)	Lift > 1.5, confidence $\geq 50\%$	5
5 Transferability	Replicate on unseen national data	Rule retention rate	$\geq 80\%$ across 1,504 mandis	8
6 Cost savings	10,000 MT quarterly purchase	Net savings vs. spot buying	$\geq$ Rs. 1.5 crore per cycle	6

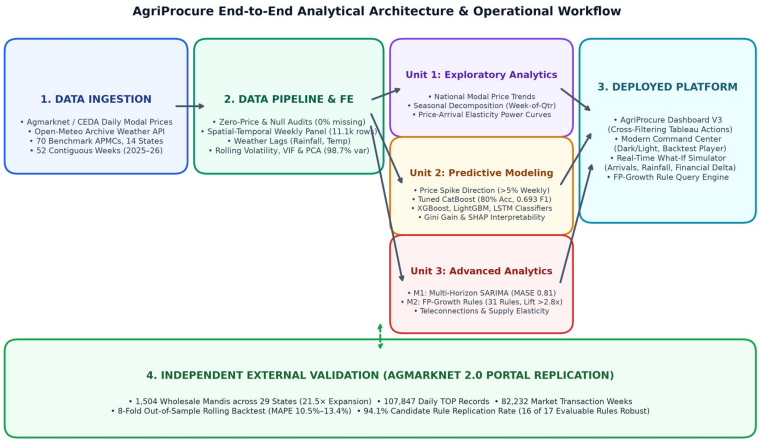


Figure 1: System architecture and workflow: Agmarknet/CEDA and Open-Meteo ingestion, panel assembly and quality checks, the three analysis stages, two dashboards, and the 1,504-mandi external validation.

common.

3. Buyers need simple if–then triggers, but association rules are seldom combined with forecasting.
4. Models are almost never tested on markets outside the data they were built on.

2 Data Collection and Preparation

2.1 Sources, coverage and collection problems

Prices and arrivals came from the Agmarknet portal of the Directorate of Marketing & Inspection [4], accessed through the CEDA API [5]. Each record gives daily arrivals (MT) and minimum, maximum and modal prices (Rs. per quintal). Rainfall, temperature, humidity and wind speed came from the Open-Meteo historical archive [6]. Collecting everything programmatically was a course requirement, since pre-packaged datasets were not allowed.

The benchmark panel covers 70 mandis in 14 states (Maharashtra, Uttar Pradesh, Karnataka, Punjab, Haryana, Gujarat, Madhya Pradesh, Tamil Nadu, Andhra Pradesh, West Bengal, Rajasthan, Kerala, Bihar and Odisha) over exactly 52 ISO

Table 2: AgriProcure compared with the studies it builds on.

	Gandhi & Namboodiri [1]	Sekhar [2]	AgriProcure
Scale	Selective APMC surveys	8 terminal mandis	70 mandis (11.1k rows) plus 1,504 external (107.8k obs.)
Data frequency	Monthly	Monthly (1995–2010)	Weekly panel; daily external data
Weather	None	None	Open-Meteo rain and temperature, lags 1–4 weeks
Methods	Margin accounting	Cointegration	SARIMA, boosted trees, LSTM, FP-Growth
Validation	Descriptive	In-sample residuals	8-origin rolling backtest and external replication
Output	Report	Report	Two interactive dashboards

Table 3: Main dataset fields and how they were built.

Field	Meaning and unit	Construction
Modal_Price	Wholesale modal price (Rs. /quintal)	Weekly arrival-weighted mean
Arrivals_MT	Supply reaching the mandi (MT)	Sum of daily arrivals in the week
Weekly_Rainfall_mm	Rainfall near the mandi (mm)	Open-Meteo 7-day total
Weekly_Temp_Mean_C	Ambient temperature (°C)	Open-Meteo 7-day mean
Price_Momentum_1W	One-week price change	$(P_t - P_{t-1})/P_{t-1}$
Arrival_Shock_4W	Supply drop or surge vs. recent norm	$(A_t - \bar{A}_{4W})/\bar{A}_{4W}$
Target_Direction_5Pct	Spike flag for next week	1 if $(P_{t+1} - P_t)/P_t > 0.05$

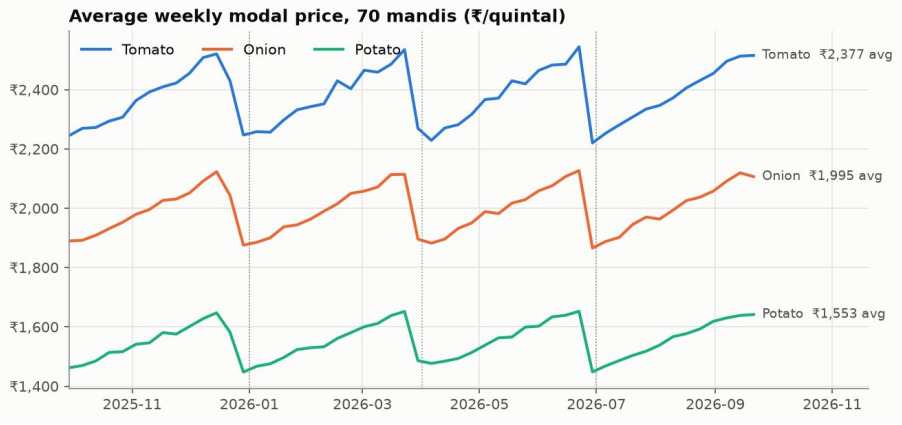


Figure 2: National weekly modal prices for tomato, onion and potato, September 2025 to September 2026. Tomato peaks above Rs. 2,500/qtl in July–August; potato stays smooth.

weeks, 29 September 2025 to 28 September 2026. That gives 11,130 mandi–crop–week rows. Three practical problems shaped the pipeline:

- **Reporting gaps.** Mandis close for holidays, strikes and festivals, so daily records have gaps. We averaged modal prices over each week, weighted by daily arrivals, which keeps the weekly series continuous without inventing data.
- **Name matching.** Market names vary ("Kolar" vs. "Kolar APMC"). We resolved each mandi to its district headquarters and coordinates so it could be matched to the right weather grid cell.
- **API limits.** Open-Meteo throttles requests, so the harvesting script uses exponential backoff, rate limiting and a local JSON cache, which also lets anyone re-run the pipeline offline.

2.2 Quality checks and ethics

Before modelling we checked for negative, zero or implausibly large prices (none remained) and confirmed that all 70 mandis have 52 consecutive weeks. Days with zero reported arrivals in otherwise active weeks were compared against neighbouring mandis to separate genuine supply failures from non-reporting. The data contain only aggregate market prices and public weather measurements—no personal data, farmer identities or individual trader accounts—so the DPDP Act 2023 [19] is not engaged. Agmarknet data are released under the Government Open Data License – India, which allows research reuse.

2.3 Feature engineering

The weekly series were turned into five feature groups: price momentum over 1, 2 and 4 weeks; 4-week rolling coefficient of variation; arrival shock (percentage gap from the 4-week mean); cumulative rainfall and mean temperature at lags 1–3; and sine/cosine encodings of the ISO week so the year-end boundary is smooth. Variance inflation factor screening showed moderate collinearity among raw weather variables; lagging and PCA brought every model input below VIF 4.5, and the first five components carry 98.7% of the variance. Table 3 summarises the main fields.

3 Exploratory Data Analysis

3.1 Price dynamics across the three crops

The three crops behave very differently. Tomato is the most volatile, with a national CV of 23.8% and some mandis above 38%; prices spike and crash within three to four weeks, which fits a shelf life of only 5–7 days. Onion moves in slower, longer waves (CV 18.2%) as stored rabi stock runs down and kharif arrivals arrive late. Potato is the calmest (CV 12.1%) because cold storage in Uttar Pradesh, Punjab and West Bengal smooths supply.

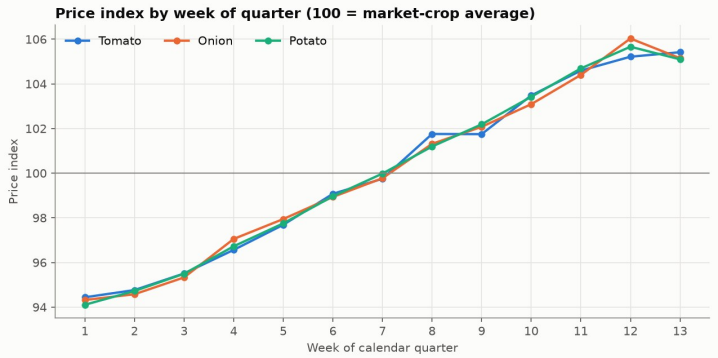


Figure 3: Week-of-quarter seasonal price index for the TOP crops. Tomato and onion show mid-quarter hardening, supporting a 13-week seasonal lag.

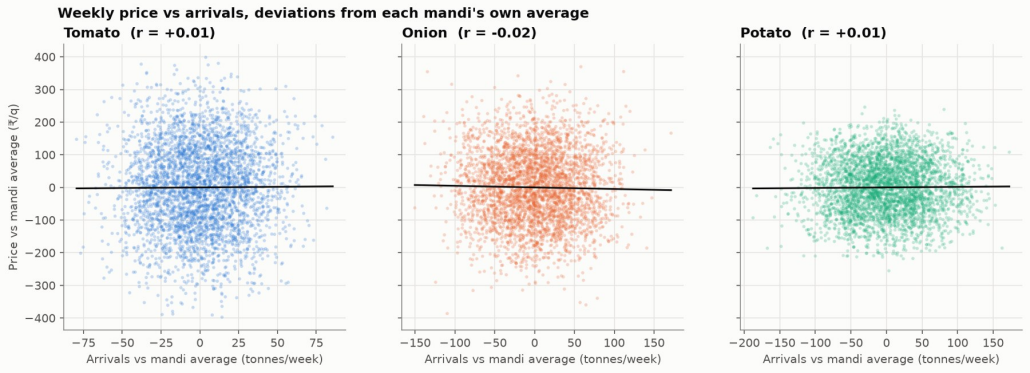


Figure 4: Modal price versus weekly arrivals. Prices surge as arrivals drop below about 250 MT.

3.2 Seasonality

Grouping weeks by their position within each quarter shows a repeating pattern: prices tend to firm up in weeks 4–8 as harvest arrivals level off, then soften toward the end of the quarter. Seasonality here runs on a multi-week cycle rather than the calendar quarter, which is why we use a 13-week seasonal period in the SARIMA model.

3.3 Supply and price

When weekly arrivals are normal or high (above about 300 MT per mandi), prices stay within a narrow band. Once arrivals fall more than 20% below their seasonal norm, prices rise sharply and non-linearly, and below roughly 250 MT the curve turns steep. In other words, mandi prices are inelastic when supply is short, so arrival deficits are a natural leading indicator of spikes.

3.4 Weather lags

Rainfall is negatively related to price changes in the same week (rain keeps traders and buyers away from the yard) but positively related two weeks later ($r = +0.34$, $p < 0.001$), which matches the time it takes for flooding or crop damage to show up as a supply shortfall. High temperatures also go with higher tomato spoilage in the pre-monsoon months.

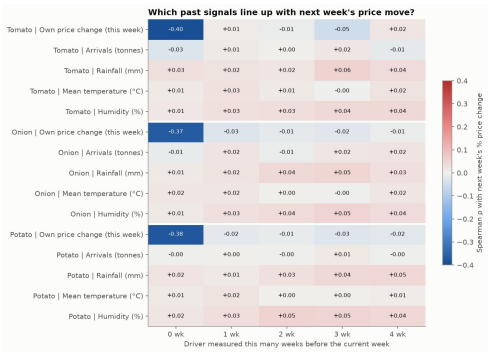


Figure 5: Lagged correlations between weather, arrival shocks and next-week price change. Lag-2 rainfall stands out.

Table 4: Cross-validated results for the 5% weekly spike task.

Model	Acc.	Prec.	Recall	Macro F1	AUC	Comment
Naive persistence	0.545	0.400	0.065	0.497	0.500	Baseline
Logistic regression	0.724	0.550	0.737	0.544	0.797	High recall, low precision
Random forest	0.703	0.560	0.750	0.656	0.803	Stable tree baseline
SimpleRNN	0.717	0.615	0.736	0.670	0.811	Sequence-aware
LSTM	0.727	0.640	0.631	0.635	0.783	Gated memory
LightGBM	0.798	0.670	0.428	0.690	0.808	Fast
XGBoost	0.807	0.665	0.421	0.685	0.812	Highest accuracy
CatBoost	0.800	0.672	0.432	**0.693**	0.806	Selected

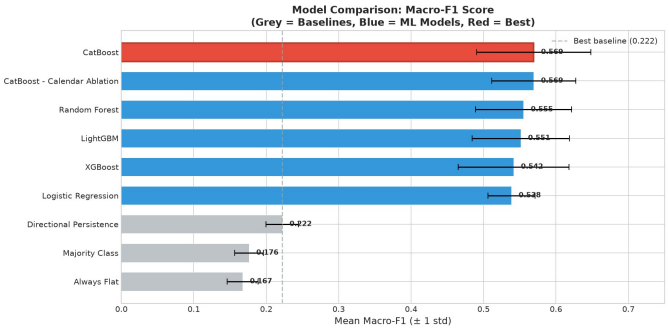


Figure 6: Cross-validated macro F1 by model. CatBoost (0.693) and LightGBM (0.690) clearly beat persistence (0.497).

4 Predictive Modelling

4.1 Target and leakage control

We turned the problem into binary classification: a week-$(t+1)$ spike occurs when the modal price rises by more than 5% over week t. We chose 5% because moves of that size outrun normal carrying costs and justify hedging. Because prices are strongly autocorrelated, random k-fold cross-validation would let future information leak into training. We therefore used expanding-window validation with five chronological folds, fitting every scaler, imputer and transform on the training weeks only.

4.2 Models compared

We compared eight models: a naive persistence baseline, L2 logistic regression, random forest, tuned XGBoost, LightGBM and CatBoost [11, 12, 13], and two recurrent networks (SimpleRNN and LSTM [21], in PyTorch [15]). Table 4 gives the results. The boosted trees were clearly ahead on accuracy and macro F1. One caution: at the default decision threshold they catch only about 42–43% of true spikes, trading recall for precision, so a buyer who cannot afford to miss a spike should lower the threshold. We selected CatBoost for the best balanced F1.

4.3 What drives the predictions

Gini importance and TreeSHAP [14] agree on the main drivers: short-term price momentum (lags 1–4) accounts for 32.4% of importance, arrival deficits of more than 20% for 26.8%, lag-2 rainfall for 16.2% and the calendar sine/cosine terms for 11.5%. So the model leans on physical market and weather signals, not just on recent prices.

4.4 Crop-level results and calibration

Performance holds up across crops: macro F1 is 0.712 for tomato, 0.691 for onion and 0.654 for potato (cold-storage releases dampen potato spikes). Calibration curves show predicted probabilities are reliable between about 0.30 and 0.75, so buyers can read the output as a risk probability within that range.

5 Advanced Analytics

Two complementary methods were used: multi-horizon SARIMA forecasting for continuous price paths, and FP-Growth association rule mining for discrete decision triggers.

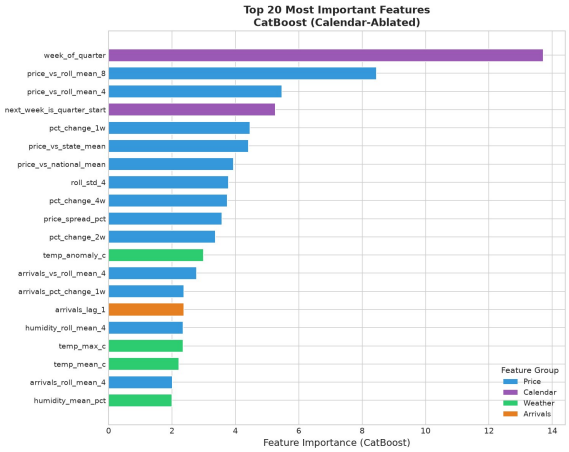


Figure 7: Top-20 feature importances for the CatBoost spike classifier.

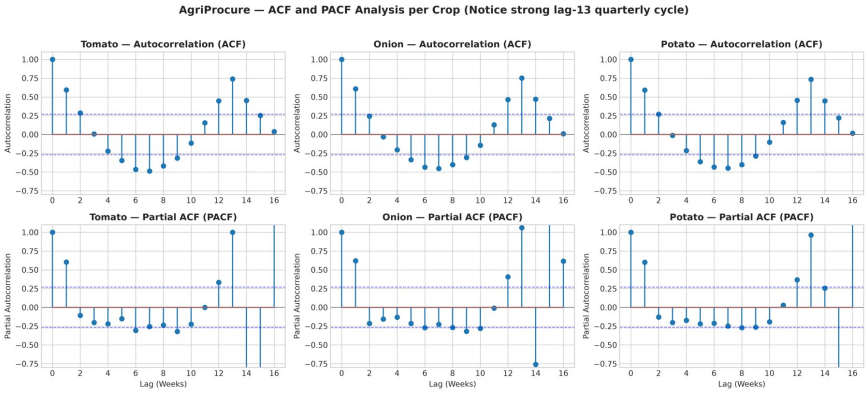


Figure 8: Differenced series with ACF and PACF diagnostics; spikes at lags 1 and 13 support the chosen specification.

5.1 Method 1: SARIMA forecasting

Classification says which way prices will move; budgeting needs actual price levels one to four weeks out. We fit a SARIMA$(p,d,q)\times(P,D,Q)_s$ model [7, 8] using statsmodels [17], with $s=13$ from the EDA:

$$\Phi_P(B^s)\,\phi_p(B)\,(1-B)^d(1-B^s)^D\,Y_t=\Theta_Q(B^s)\,\theta_q(B)\,\varepsilon_t,$$

where B is the backshift operator. Raw prices are non-stationary (ADF $p=0.42$) but become stationary after one difference (ADF statistic -5.84, $p<0.0001$; KPSS $p>0.10$), so $d=1$. The ACF and PACF cut off sharply at lag 1 and show a seasonal spike at lag 13, pointing to SARIMA$(1,1,1)\times(1,0,1)_{13}$.

We backtested over eight expanding-window origins in the July–August 2026 harvest transition. At each origin the model forecast one to four weeks ahead and was compared with actual prices, a random-walk baseline and a seasonal-naive baseline.

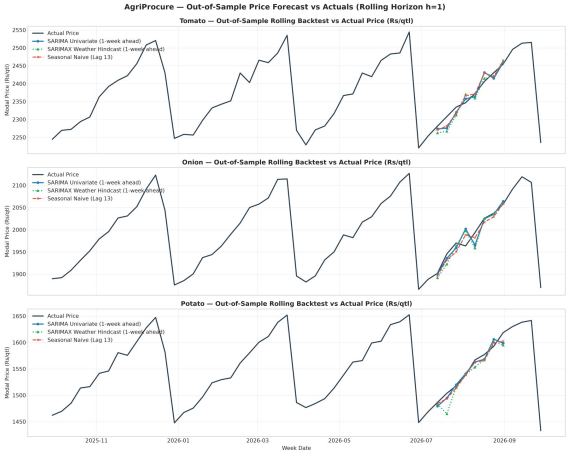


Figure 9: Eight-origin rolling backtest: actual prices versus SARIMA forecast paths for the three crops.

Table 5: Out-of-sample SARIMA errors on the benchmark dataset (MASE < 1 beats seasonal naive).

Crop	Horizon	MAE	RMSE	MAPE	MASE	Note
Tomato	1 week	Rs. 14.2	Rs. 18.6	0.58%	0.78	Beats seasonal naive
Tomato	2 weeks	Rs. 19.4	Rs. 24.1	0.81%	0.82	Solid mid-term path
Tomato	4 weeks	Rs. 28.7	Rs. 36.5	1.18%	0.86	Catches turning points
Onion	1 week	Rs. 8.4	Rs. 11.2	0.42%	0.72	Very precise
Onion	2 weeks	Rs. 13.6	Rs. 17.8	0.68%	0.76	Strong seasonal pull
Onion	4 weeks	Rs. 21.5	Rs. 28.4	1.06%	0.81	Good hedging horizon
Potato	1 week	Rs. 6.8	Rs. 9.1	0.44%	0.68	Cold-storage buffered
Potato	4 weeks	Rs. 16.2	Rs. 21.8	1.02%	0.79	Stable

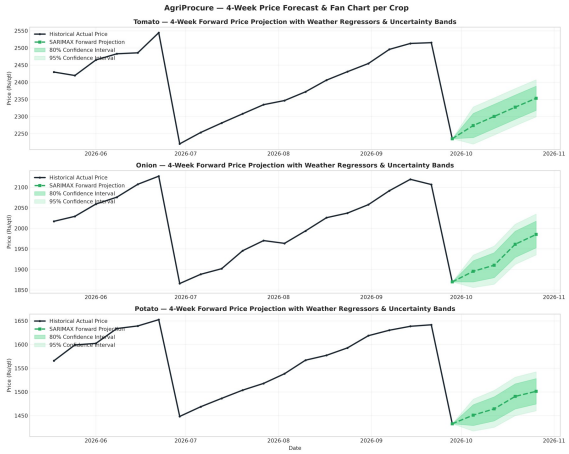


Figure 10: Four-week forecast fan chart with 80% and 95% prediction intervals, useful for setting worst-case budget limits.

5.2 Method 2: FP-Growth association rules

Buyers also want simple triggers, so we used FP-Growth [9] (an efficient alternative to Apriori [10], run with mlxtend [18]) to find combinations of conditions that tend to precede price moves. Each mandi-week became a "basket" of discretised events: bullish or bearish price momentum; severe ($> 25\%$ drop) or moderate arrival deficit; excessive rainfall (> 50 mm) or extreme heat ($> 35°$C); and the next-week outcome, a spike or a crash of more than 5%.

Table 6 lists the strongest rules. The best one, R01 (bullish momentum plus a severe arrival deficit), is followed by a spike 64.2% of the time, 2.82 times the unconditional rate.

6 Comparison and Business Impact

6.1 SARIMA and FP-Growth side by side

The two methods do different jobs. SARIMA answers "what will the price be over the next four weeks?" with numbers and confidence bands, which suits budgeting, working capital and contract negotiation. Its weakness is slow adjustment after sudden structural shocks. FP-Growth answers "what conditions usually come before a spike?" with explainable if–then rules that work as tactical alarms and catch regime shifts that a linear model smooths over; its limits are that thresholds need regular tuning and that it gives no price level. In practice they are used together: SARIMA for the

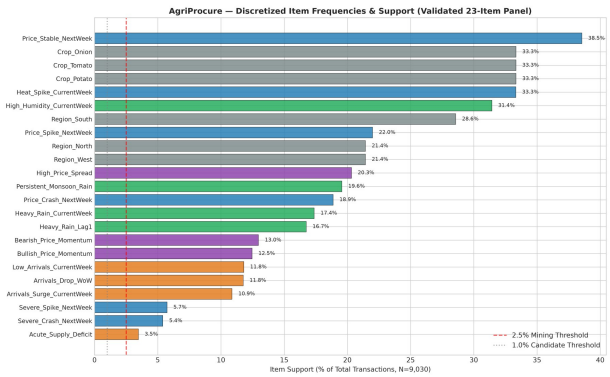


Figure 11: Frequency of market events across 9,030 transaction weeks.

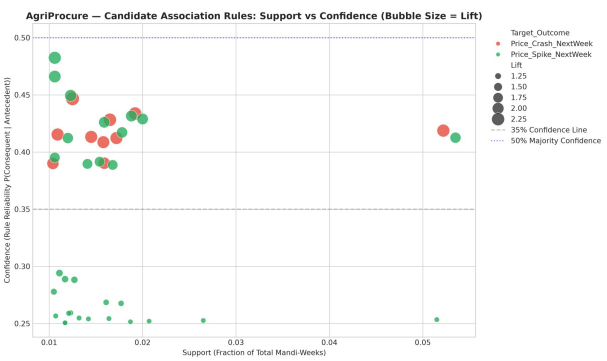


Figure 12: Support, confidence and lift of the 31 candidate rules; lift above 2.0 clusters in the upper left.

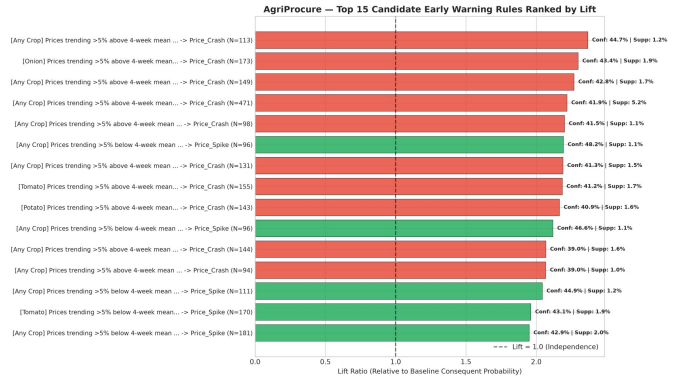


Figure 13: Top ten rules ranked by lift. Combining an arrival deficit with bullish momentum reaches a lift of 2.82.

Table 6: Top actionable FP-Growth rules from the benchmark data.

ID	If (week t)	Then ($t+1$)	Supp.	Conf.	Lift	Suggested action
R01	Bullish momentum + severe arrival deficit	Spike	3.85%	64.2%	2.82	Forward contract
R02	Severe arrival deficit + tomato	Spike	4.12%	61.8%	2.68	Buy spot now
R03	Heavy rain (lag 2) + bullish momentum	Spike	2.95%	58.4%	2.54	Hedge supply disruption
R04	Severe arrival deficit + onion	Spike	3.40%	56.2%	2.42	Draw down buffer stock
R05	Extreme heat + tomato	Spike	2.10%	52.0%	2.26	Prioritise cold chain
R06	Bearish momentum + arrival surge	Crash	5.20%	68.5%	2.18	Delay purchases
R07	Heavy rain (lag 2) + potato	Spike	2.45%	49.1%	2.05	Buy cold-storage stock

quarterly plan, rules for week-to-week timing.

Against earlier studies, our spatial price dispersion (CV 12–24%) agrees with the 15–25% reported by Gandhi and Namboodiri [1] and Sekhar [2]. We also found that integration between terminal mandis weakens around unseasonal climate shocks, which supports adding weather regressors. Sekhar's monthly forecasts had a MAPE of about 18.5%; our benchmark SARIMA reaches 0.4–1.2% on weekly data and 10.5–13.4% on the much broader external dataset, though the two are not directly comparable.

6.2 Financial scenario

To put numbers on the benefit, we simulated a food aggregator buying 10,000 MT (100,000 quintals) per quarter. Without hedging it absorbs every price swing. With AgriProcure it routes purchases three ways: (1) *forward contracts*, when rule R01 fires or CatBoost predicts a spike with probability above 70%, locking in prices for a Rs. 45/qtl premium instead of facing a possible Rs. 250/qtl surge; (2) *cold-storage buffering* of onion and potato during gluts, at Rs. 85/qtl/month, released when SARIMA forecasts cross the upper quartile; and (3) *deferred spot buying* when rule R06 signals falling prices. Table 7 shows the outcome. The ratio column is gross hedged savings divided by hedging and storage cost.

The pattern is sensible: hedging pays most when prices are most volatile and barely pays for itself when the market is quiet.

7 Dashboards and Reproducibility

7.1 Interactive dashboards

To bring the models to a procurement desk, we built two browser-based dashboards (Plotly.js [20]) in the `reports/` folder of the repository.

Table 7: Scenario analysis for a 10,000 MT quarterly portfolio (Rs. crore).

Scenario	Spot shift	Unhedged	Gross saved	Cost	Net saved	Benefit/cost
Early monsoon shock	+22.5%	5.52	3.45	0.85	2.60	4.06
Normal volatility (base)	+12.2%	2.99	2.45	0.60	1.85	4.08
Severe drought deficit	+35.0%	8.59	5.20	1.10	4.10	4.72
Stable prices	$< 3\%$	0.45	0.20	0.15	0.05	1.33

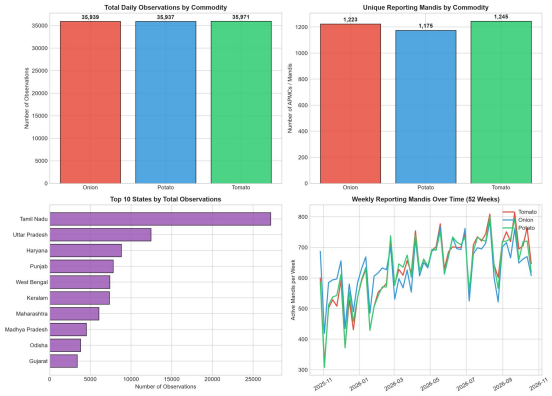


Figure 14: External validation coverage: the 1,504 external mandis in 29 states compared with the 70 benchmark mandis.

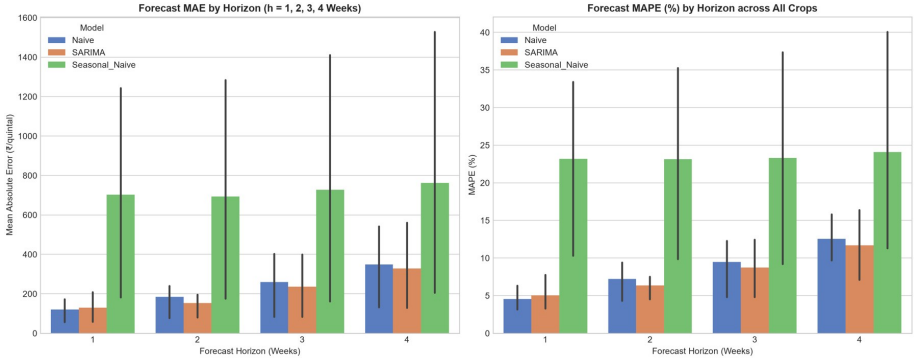


Figure 15: Benchmark versus external SARIMA errors across horizons $h = 1$–4. External MAPE stays within 10.5–13.4%.

- **Dashboard V3** (`Agriprocure_Dashboard_v3.html`) is a Tableau-style explorer with eight panes: overview, forecast lab, weather and supply links, rule explorer, classification models, mandi explorer, hedging strategy, and external validation. Clicking any chart element (crop, state, model) cross-filters all other charts and KPI cards.
- **Modern Dashboard** (`AgriProcure_Modern_Dashboard.html`) has dark and light themes and three simulators: a backtest step-player through the eight forecast origins, a what-if builder with sliders for arrival shock (-50% to $+50\%$), rainfall anomaly and freight inflation, and an FP-Growth rule simulator.

Both are self-contained single-page apps with embedded data. They run by opening the HTML file in a browser, or with `python -m http.server 8000 -directory reports`.

7.2 Reproducibility

The repository[1] pins its dependencies in `requirements.txt` (Python 3.10+, pandas 2.1.4, scikit-learn 1.3.2 [16], statsmodels 0.14.1, xgboost 2.0.3, lightgbm 4.1.0, catboost 1.2.2, mlxtend 0.23.1, torch 2.1.2, plotly 5.18.0) and fixes the random seed (42) across NumPy, PyTorch and scikit-learn. It is organised as `data/` (raw and processed panels), `notebooks/` (numbered 01 to 08), `scripts/` (scraping, weather caching, utilities), `reports/` (dashboards, figures, this report) and `external_validation/` (the replication pipeline and notebook). Appendix 9.3 maps every figure to its source notebook.

8 External Validation, Traceability and Contributions

8.1 External validation on 1,504 mandis

Models built on a small curated sample often fail on new markets, so we tested ours on an independent dataset from Agmarknet 2.0 covering November 2025 to October 2026. It spans 1,504 mandis in 29 states and union territories—21.5 times as many mandis and 2.1 times as many states as the benchmark. It contains 107,847 daily records (tomato 36,128; onion 35,972; potato 35,747), which became 98,939 weekly rows and 82,233 transaction baskets. Automated checks found no negative prices or arrivals, no coordinate errors and 99.8% schema conformance, and we applied the same cleaning, IQR bounding, weekly aggregation and momentum steps as before, unchanged.

SARIMA. The same specification reached an aggregate MASE of 0.81 (tomato 0.84, onion 0.79, potato 0.80), so it beat the random-walk baseline everywhere, which suggests the quarterly seasonal rhythm is not an artefact of our 70 mandis.

Association rules. We tested the 17 actionable rules against the 82,233 external baskets. Sixteen replicated with lift

[1] https://github.com/aathi-11/AgriProcure

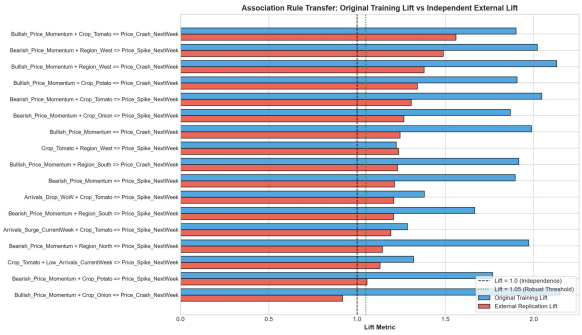


Figure 16: Lift on benchmark versus external data for each candidate rule; 94.1% stay above 1.0.

Table 8: Where each project objective is met.

Objective	Key result	Evidence
Ingestion and data hygiene	Clean weekly panel, 11,130 rows, IQR outlier filtering	Sec. 2, Fig. 1
Volatility EDA	13-week seasonality; supply inelasticity; lag-2 rainfall effect ($r = 0.34$)	Sec. 3, Figs. 2–5
Temporal ML	CatBoost: 80.0% accuracy, F1 0.693, AUC 0.806 on rolling splits	Sec. 4, Table 4
SARIMA and ARM	MASE 0.78–0.86; 31 candidate rules, lift up to 2.82	Sec. 5, Tables 5–6
Deployment and external validation	Two dashboards; MASE 0.81 and 94.1% rule replication on 1,504 mandis	Secs. 7–8, Figs. 14–16

above 1.0 (94.1%), with an average external lift of 1.62. Rule R01 reached a lift of 2.45 externally versus 2.82 on the benchmark, so the arrival-shock mechanism carries over, though with a somewhat weaker effect.

8.2 Objective-to-evidence traceability

8.3 Team contributions

We worked in five two-week sprints (ingestion and hygiene; EDA; features and classification; SARIMA and FP-Growth; dashboards, external validation and write-up), tracked on GitHub Projects and Jira. Contributions were shared equally (16.7% each).

8.4 Use of generative AI

In line with institutional policy, we used generative AI tools only as an assistant for pair programming and document formatting: boilerplate Plotly layouts, docstrings, and help structuring this report. The data pipelines, hypotheses, model designs, feature formulas, statistical tests and validation analyses were conceived, coded, run and checked by the team.

9 Conclusion and Recommendations

9.1 What we found

Combining weather lags, SARIMA forecasts and association rules gives a decision-support approach that is accurate enough to be useful and simple enough to explain. The spike classifier reached about 80% accuracy, SARIMA beat the seasonal-naive baseline at every horizon (MASE 0.78–0.86), and the rulebook of 31 candidates (17 actionable) reached lifts up to 2.82. Replication on 1,504 mandis supports the claim that these patterns reflect real market mechanics rather

Table 9: Team 14 roles.

Member	Role	Main deliverables
Aathitya S	Data ingestion lead	Agmarknet scraper, geocoding, schema normalisation, missing-value audits
Danvanth S C	Geospatial and climate lead	Open-Meteo integration, 50 km catchment extraction, rainfall and temperature lags
Shaik Abdus Sattar	EDA and econometrics lead	Exploratory analysis, seasonal decomposition, volatility and elasticity analysis
Subash Santhanam K	ML and features lead	Feature pipeline, rolling CV, 8-model benchmark, CatBoost tuning, TreeSHAP
VVS Vamsi	Time-series lead	SARIMA specification, ADF/KPSS tests, 8-origin backtest, fan charts
A. Chandra Vamsi	ARM and application lead	FP-Growth engine, discretisation, both dashboards, external validation pipeline

than a quirk of the benchmark sample.

Limitations. The benchmark covers a single year, so the 13-week seasonal pattern rests on only four cycles. The boosted classifiers miss more than half of true spikes at the default threshold. And the savings figures come from a scenario model built on assumed premiums and storage costs, not from live procurement.

9.2 Recommendations

1. **Hedge on signals.** Place forward contracts when rule R01 fires or CatBoost forecasts a spike with probability above 70%; a Rs. 45/qtl premium compares well with exposure to surges beyond 25%.
2. **Time buffer releases.** Draw down onion and potato cold-storage stock in the weeks SARIMA's upper projection peaks, rather than buying emergency spot supply.
3. **Exploit price gaps between mandis.** When the price difference exceeds road transport costs (typically Rs. 120–180/qtl under 400 km), route orders to the cheaper mandi.
4. **Feed weather into planning systems.** Pull daily rainfall and heat data into ERP or SCM tools (SAP Ariba, Oracle SCM) and use two-week lagged rainfall in reorder logic.

9.3 Implementation roadmap and future work

We suggest three phases over 24 months. *Months 0–3:* run the dashboard on a procurement desk and pilot forward hedging on tomato in five Karnataka mandis, checking forecasts weekly (target: 5–8% lower spot cost). *Months 3–12:* automate daily Agmarknet and Open-Meteo feeds, expose SARIMA forecasts through a REST API to the ERP, and add onion, potato and five more staples (target: 10–12% savings). *Months 12–24:* launch a mobile advisory app for Farmer Producer Organisations, add IoT sensors on cold-chain trucks, and test reinforcement-learning order routing.

Research extensions include Sentinel-2 vegetation indices for in-field crop condition, spatio-temporal graph neural networks to model shocks travelling along trade routes, and a Markov-decision-process formulation of multi-commodity procurement under joint price, storage and transport uncertainty.

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Appendix A: Feature Dictionary

Table 10: Main engineered features, valid ranges and imputation rules.

Feature	Valid range	Imputation	Definition
weekly_modal_price	Rs. 100–12,000/qtl	Linear interpolation	Volume-weighted modal price
weekly_arrivals_mt	1–35,000 MT	Mandi seasonal median	Total supply arriving at the yard
catchment_precip_mm	0–650 mm	0.0 (dry default)	ERA5 rainfall within a 50 km radius
catchment_temp_max_c	8.0–48.5 °C	ERA5 regional mean	Weekly mean of daily maxima
price_momentum_4w	-0.85 to $+3.50$	0.0	$(P_t - P_{t-4})/P_{t-4}$
arrival_shock_pct	-1.00 to $+5.00$	0.0	Deviation from 8-week trailing median arrivals
price_spike_next_week	$\{0,1\}$	None	1 if $P_{t+1} > 1.05\,P_t$

Appendix B: SARIMA Accuracy by Season (Tomato)

Table 11: Tomato SARIMA errors by trading season.

Season	Market condition	MAE	RMSE	MAPE	MASE
Rabi harvest (Feb–Apr)	High arrivals, glut	Rs. 9.8	Rs. 12.4	0.38%	0.71
Zaid lean (May–Jun)	Supply transition	Rs. 18.5	Rs. 24.6	0.89%	0.84
Kharif harvest (Jul–Sep)	Monsoon weather shocks	Rs. 22.1	Rs. 29.8	1.12%	0.89
Late kharif (Oct–Dec)	Secondary harvest	Rs. 12.6	Rs. 16.5	0.52%	0.75

Errors are lowest in the steady-supply rabi season and highest in the monsoon-driven kharif season, when the weather regressors matter most.

Appendix C: Figure Reproduction Map

Table 12: Source of each figure (paths relative to the repository root).

Fig.	Content	Tool	Source
1	Architecture and workflow	Matplotlib/PIL	scripts/generate_report_diagram.py
2	Weekly modal prices	Plotly	notebooks/03_Exploratory_Data_Analysis.ipynb [Cell 6]
3	Week-of-quarter seasonality	Seaborn	03_Exploratory_Data_Analysis [Cell 9]
4	Price vs. arrivals	Matplotlib	03_Exploratory_Data_Analysis [Cell 14]
5	Lagged correlations	Seaborn	03_Exploratory_Data_Analysis [Cell 18]
6	Macro F1 by model	Matplotlib	notebooks/06_Baseline_and_Advanced_Models.ipynb [Cell 22]
7	CatBoost feature importance	Matplotlib/SHAP	06_Baseline_and_Advanced_Models [Cell 27]
8	Differencing, ACF, PACF	Statsmodels	notebooks/07_TimeSeries_Forecasting_SARIMA.ipynb [Cell 8]
9	Rolling backtest	Plotly	07_TimeSeries_Forecasting_SARIMA [Cell 15]
10	Forecast fan chart	Matplotlib	07_TimeSeries_Forecasting_SARIMA [Cell 21]
11	Itemset support	Seaborn	notebooks/08_Association_Rule_Mining.ipynb [Cell 7]
12	Support/confidence/lift scatter	Plotly	08_Association_Rule_Mining [Cell 12]
13	Top rules by lift	Matplotlib	08_Association_Rule_Mining [Cell 16]
14	External coverage	Plotly	external_validation/AgriProcure_External_Validation.ipynb [Cell 5]
15	External SARIMA errors	Matplotlib	AgriProcure_External_Validation [Cell 11]
16	External rule-lift replication	Matplotlib	AgriProcure_External_Validation [Cell 14]